



Bahria University, Islamabad

Department of Software Engineering

Artificial Intelligence - Lab (Spring-2026)

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Lab Journal: 03

Date: 02/20/2026

Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

OOP in Python (Part-4) (Functions, Classes, Inheritance, Polymorphism)

Task1:

Create a base class Shape with an abstract method calculate_area(). Implement the calculate_area() method in the following subclasses:

- Rectangle (Length × Width)
- Square (Side²)
- Circle ($\pi \times \text{Radius}^2$)
- Cylinder ($2\pi r(r + h)$)

Test your classes by creating objects for each shape and calculating their areas.

Code:

```
from abc import ABC, abstractmethod
import math

class Shape(ABC):

    def __init__(self, name):
        self.name = name

    @abstractmethod
    def calculate_area(self):
        pass

    def display_info(self):
        print(f"{self.name} - Area: {self.calculate_area():.2f}")

class Rectangle(Shape):
    def __init__(self, length, width):
        super().__init__("Rectangle")
        self.length = length
        self.width = width

    def calculate_area(self):
        return self.length * self.width

class Square(Shape):
    def __init__(self, side):
```

```
        super().__init__("Square")
        self.side = side

    def calculate_area(self):
        return self.side ** 2

class Circle(Shape):
    def __init__(self, radius):
        super().__init__("Circle")
        self.radius = radius

    def calculate_area(self):
        return math.pi * self.radius ** 2

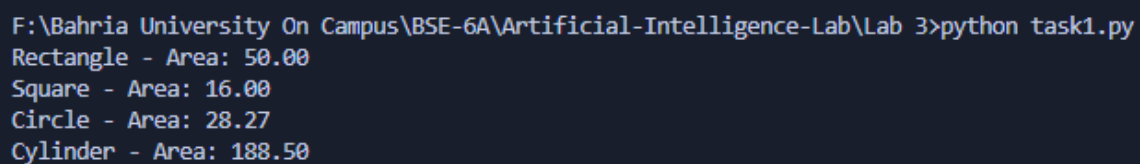
class Cylinder(Shape):
    def __init__(self, radius, height):
        super().__init__("Cylinder")
        self.radius = radius
        self.height = height

    def calculate_area(self):
        # Surface area: 2πr(r + h)
        return 2 * math.pi * self.radius * (self.radius +
self.height)

# Test the classes
if __name__ == "__main__":
    rect = Rectangle(10, 5)
    square = Square(4)
    circle = Circle(3)
    cylinder = Cylinder(3, 7)

    rect.display_info()
    square.display_info()
    circle.display_info()
    cylinder.display_info()
```

Screenshot



```
F:\Bahria University On Campus\BSE-6A\Artificial-Intelligence-Lab\Lab 3>python task1.py
Rectangle - Area: 50.00
Square - Area: 16.00
Circle - Area: 28.27
Cylinder - Area: 188.50
```

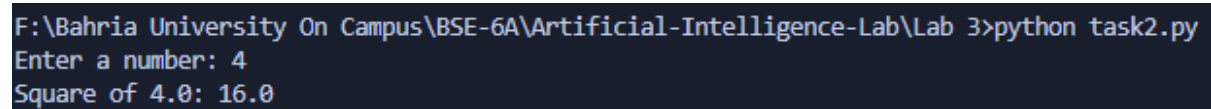
Tasks 2 : Implement a lambda expression to calculate square of given number.

Code:

```
square = lambda x: x ** 2

if __name__ == "__main__":
    number = float(input("Enter a number: "))
    result = square(number)
    print(f"Square of {number}: {result}")
```

Screenshot



```
F:\Bahria University On Campus\BSE-6A\Artificial-Intelligence-Lab\Lab 3>python task2.py
Enter a number: 4
Square of 4.0: 16.0
```

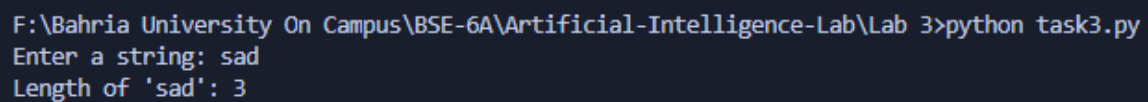
Task 3: Implement a lambda expression to calculate the length of passed string.

Code

```
# this is lamda expression to calculate lenght of string
string_length = lambda s: len(s)

# Take input from user
if __name__ == "__main__":
    text = input("Enter a string: ")
    result = string_length(text)
    print(f"Length of '{text}': {result}")
```

Screenshot



```
F:\Bahria University On Campus\BSE-6A\Artificial-Intelligence-Lab\Lab 3>python task3.py
Enter a string: sad
Length of 'sad': 3
```

Task 4: Create a generator function that returns 5 random numbers between 1 and 100. Use a `for` loop to print the numbers

Code:

```
import random
def random_numbers():
    for _ in range(5):
        yield random.randint(1, 100) # return the
if __name__ == "__main__":
    print("5 random numbers between 1 and 100:")
    for number in random_numbers():
        print(number)
```

Screenshot

```
F:\Bahria University On Campus\BSE-6A\Artificial-Intelligence-Lab\Lab 3>python task4.py
5 random numbers between 1 and 100:
53
67
95
52
41
```

Task 5: Implement a generator function that returns the Fibonacci sequence up to a given limit **n**. Use a **for** loop to print the sequence.

Code

```
def fibonacci(n):
    a, b = 0, 1
    while a <= n:
        yield a
        a, b = b, a + b

if __name__ == "__main__":
    limit = int(input("Enter the limit for Fibonacci sequence: "))
    print(f"Fibonacci sequence up to {limit}:")
    for number in fibonacci(limit):
        print(number)
```

Screenshot

```
F:\Bahria University On Campus\BSE-6A\Artificial-Intelligence-Lab\Lab 3>python task5.py
Enter the limit for Fibonacci sequence: 5
Fibonacci sequence up to 5:
0
1
1
2
3
5
```